

Case Study: Vessel Valuation Tool

Scenario

You are a Finance Solution Developer working in the Business Intelligence team at a global shipping company. Management is seeking a tool to evaluate potential vessel investments. Your task is to design and build a lightweight prototype application that allows users to:

- Input key financial assumptions
- Calculate vessel lifetime returns
- Generate insights for decision-making

You've been provided with a simplified model that includes the following fields:

- Vessel Purchase Price
- Vessel Expect Life (in years)
- Vessel Residual Value
- Revenue per Day
- Off Hire Rate (%)
- Operating Expense (OpEx) per Day
- Drydock CapEx Cost
- Upgrades CapEx Cost
- Inflation Rate Assumption
- Discount Rate Assumption

Note : All data provided in this case are entirely hypothetical and created solely for assessment purposes.

1. **Offhire** refers to periods when the vessel is not moving due to technical or other issues. During the Offhire period, the vessel does **Not generating Revenue**, but it **still incurs Operating Expense**
2. **Drydock** refers to the process of removing a vessel from water for maintenance, repairs, and inspection, usually happen **every 5 years** during the vessel's useful life. Drydock costs are treated as **CapEx**.
3. **Upgrades** refers to improvements or installation of new equipment and are also treated as **CapEx**.
4. **Inflation** should be applied to **cost item only**, including OpEx and CapEx, but **Not to Revenue** as revenue is assumed to be fixed under contract.

Your task is to develop a small financial application (Python-based or similar) that:

1. Captures key vessel investment assumptions in input fields
2. Computes core financial metrics, such as Cashflows, Net Present Value (NPV), Internal Rate of Return (IRR) using the [Sample Data for Testing]
**Note: Sample Data for Testing could have errors, please have a data clean processing based on your justification*
3. Provides **decision-support insights**, such as:
 - Minimum required Revenue per Day to achieve \$0 NPV
 - Sensitivity analysis showing how changes in Revenue per Day, in increments of \$1,000, impact IRR
 - Scenario analysis showing the return for Best/Base/Worst Case when inflation assumption change

First Round: Deliverables

Please submit the following:

1. Working Application Code

Jupyter Notebook **OR** simple application (Dash preferred but not required)

Please include:

- An input section with editable assumptions
- Clearly structured calculation logic
- An output section with key metrics and insights

2. Code Quality

- Clear function structure such as `calculate_irr()`, `calculate_npv()`
- Comments explaining your logic
- Clean and readable design

3. Short Write-up (maximum 1 page)

- Your design approach
- Key assumptions made
- How your solution can scale to production use

Second Round: Interview Presentation

If selected for the next stage, you will be invited to present your solution during a 20-minute session with the hiring team.

Please be prepared to walk us through:

- Your thought process
- Visuals and structure of the dashboard
- Key insights and recommendations for senior leadership

The interview will include questions to assess your business understanding, communication skills, and technical reasoning.

Reference Formulas

To support your analysis and visualization, please refer to the simplified formulas and business logic used in this case:

- **Daily Revenue** = Revenue per Day * (1 – Off Hire Rate %)
- **Annual Revenue** = Daily Revenue * 365
- **Inflation Factor in Year t** = (1 + Inflation Rate) ^ t
- **Annual OpEx in Year t** = OpEx per Day * 365 * Inflation Factor in Year t
- **Annual CapEx in Year t** = Drydock CapEx in Year t + Upgrade CapEx in Year t
- **Annual Free Cashflow in Year t** = Annual Revenue – Annual OpEx – Annual CapEx

- **Net Present Value (NPV)**

$$NPV = \sum_{t=0}^T \frac{CF_t}{(1+r)^t}$$

- **Internal Rate of Return (IRR)**

$$0 = \sum_{t=0}^T \frac{CF_t}{(1+IRR)^t}$$

Where:

- CF_t = Net cashflow at year t
- r = Discount rate
- T = Vessel expected life (in years)

Cashflow Timing Convention

As simplified in the interview case study, all cashflows in this case should be assumed to occur at the **end of each year (December 31)**.

- **Year 0:** Initial vessel purchase investment
- **Year 1 to Year T:** Annual operating cashflows
- **Final Year (Year T):** Annual operating cashflow plus vessel residual value

Additional Guidance

Please clearly state any additional assumptions you make where the case does not provide explicit detail.

We are not only assessing whether the model works, but also how clearly you structure your logic, communicate your thinking, and design a solution.